

A generic NOVA pass separates a DNT training set's actives from its negative

EPA/Shafer multi-well MEA network formation assay (Brown et al. 2016)
— exploratory reanalysis

2026-07-15

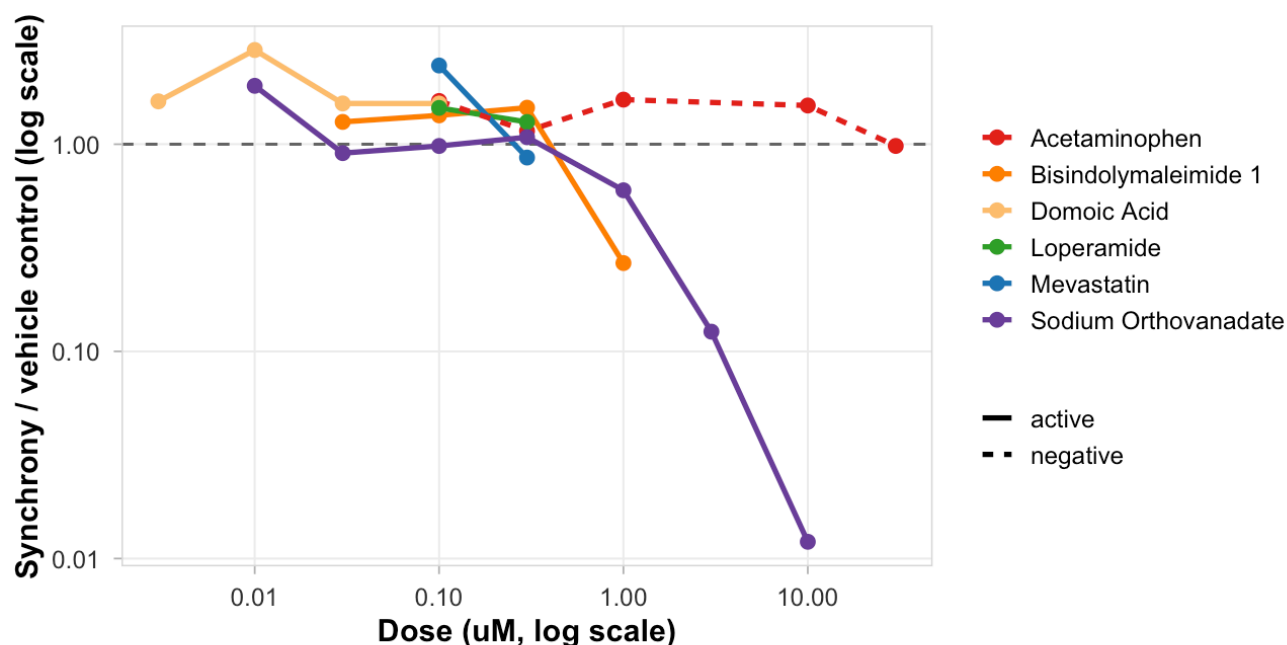
Headline

NOVA's standard pipeline was run over this dataset with no compound-specific tuning: normalise to same-plate vehicle controls, PCA, trajectory summary, rank by effect size. Brown et al. report that every compound in this training set except acetaminophen produced concentration-related effects.

All 25 of the 25 largest changes are among the paper's five actives, and acetaminophen appears 0 times. The result does not depend on the curation rule below: on the filtered ranking it is 25 of 25 actives and 0 acetaminophen. 24 of the raw top 25 are decreases relative to vehicle.

This is a statement about **prediction, not neurotoxicity**, and a weaker one than it first looks (see the caveats): 20 of the raw top 25 are compounds the paper reports as cytotoxic, and the viability data that would separate "changed the network" from "killed the cells" is not in this file.

Hero figure — the negative is flat, the actives are not



Lines join doses of one compound; no dose-response model was fitted.

Figure 1: Synchrony index as a fraction of same-plate vehicle controls at DIV 7, for every compound with at least 4 replicate wells at a given dose. Dashed line = vehicle. Points are the condition median across replicate wells, counted by plate and well ID. Sodium orthovanadate falls monotonically to ~1% of vehicle. Acetaminophen – the paper's only negative – shows no dose trend, though it sits above vehicle throughout. Mevastatin appears as two points only: the $n \geq 4$ rule removes its higher doses, so this panel understates it.

The largest changes that more than one well agrees on

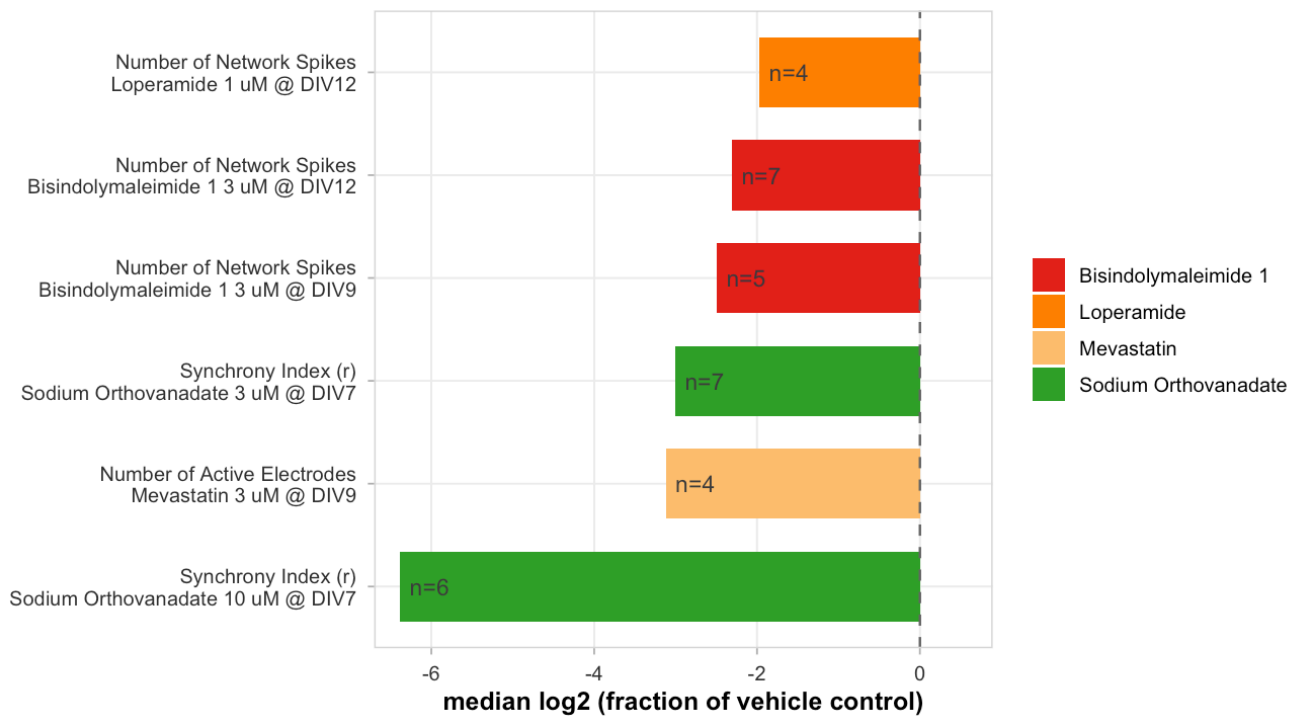


Figure 2: Top changes with at least 4 replicate wells, selected by effect-size rank from findings.csv rather than by eye. Bars are median log2 fold-change against same-plate vehicle controls – not against a baseline timepoint. Well counts are printed on each bar.

Condition	Metric	DIV	vs control	Wells	Cultures
Sodium Orthovanadate 10 uM	Synchrony Index (r)	DIV7	83.44x down	6	6
Mevastatin 3 uM	Number of Active Electrodes	DIV9	8.70x down	4	4
Sodium Orthovanadate 3 uM	Synchrony Index (r)	DIV7	8.03x down	7	6
Bisindolymaleimide 1 3 uM	Number of Network Spikes	DIV9	5.64x down	5	4
Bisindolymaleimide 1 3 uM	Number of Network Spikes	DIV12	4.96x down	7	6
Loperamide 1 uM	Number of Network Spikes	DIV12	3.92x down	4	4

Caveats

⚠ Warning

Exploratory. Nothing here is a tested claim.

- **The ranking is dominated by compounds the source reports as cytotoxic.** 20 of the raw top 25 are Bis-1, mevastatin or loperamide. Brown et al. single out **domoic acid and sodium orthovanadate** as the two that “altered several of these parameters in the absence of cytotoxicity” — the implication being that the others’ effects are at least partly cell death. The Cell Titer Blue viability data is **not in this file**, so this analysis cannot separate “changed the network” from “killed the cells”. “Recovers the training set’s actives” is therefore partly “recovers cytotoxicity”.
- **Fully-silenced wells are absent from the ranking.** A log-ratio cannot represent zero, so 1053 observations with a value of exactly 0 vs control are excluded — and those are the *largest possible* effects. 4 conditions disappear entirely as a result (Loperamide 10 uM; Loperamide 30 uM; Mevastatin 30 uM; Bisindolymaleimide 1 10 uM), which are top doses of three actives. The ranking therefore systematically under-represents complete network silencing. This is also why the trajectory section counts 45 conditions while the ranking sees 41.
- **Selection inflates the top of the list.** 944 condition × metric × timepoint comparisons were scanned and sorted by effect size; sorting a large family of noisy estimates puts the luckiest first. No correction is applied because nothing is a hypothesis test.
- **199 of 944 comparisons rest on fewer than 4 wells** and are excluded from the figures and table. Raw effect size cannot distinguish a large change in three wells from one in thirty: unfiltered, 14 of the top 25 rest on a **single well**. The largest single-well effect is 213x; the largest effect overall is 1680x, on 3 wells.
- **DIV 5 is excluded entirely.** Values are a fraction of same-plate vehicle controls, and at DIV 5 those controls are themselves silent: the control mean is exactly zero for network spikes in **5 of 6 experiments**, leaving the ratio undefined for **63–81% of wells on the network metrics** (0% on firing rate, active electrodes and synchrony). Cortical networks have not formed at DIV 5; that is what this assay measures.
- **No timepoint here is pre-treatment.** Compounds are applied from the start of the 12-day culture, so “change across DIV” is not “onset of effect”.
- **Plate, batch and dose structure are not modelled.** Wells nest in cultures, cultures reuse physical plates, doses nest in compounds, and DIV is a repeated measure on the same wells. Conditions are treated as flat groups.
- **Prediction is not causation.** Recovering the training set’s own active/inactive labels says the assay predicts those labels. It says nothing about developmental neurotoxicity in humans.
- **The burst-interval and network-spike *shape* metrics are excluded from the state space** (mean.isis, mean.dur, mean.IBIs, ns.peak.m, ns.durn.m, ns.mean.insis, ns.durn.sd, ns.mean.spikes.in.ns), because their 33–42% missingness means “this well did not burst” rather than “measurement lost”, and mean-imputing would invent burst statistics for silent wells. The network-spike **counts** (ns.n, ns.percent.of.spikes.in.ns) have no missingness and *are* in the state space.

Source

US EPA (Brown et al. 2016, *Toxicol Sci* 154(1):126-139, doi:10.1093/toxsci/kfw147), public domain (17 USC 105). URL, SHA-256 checksum, the compound designations quoted from the abstract, and the full structural record are in `SOURCE.md`, reproduced in the HTML notebook.

Generated with NOVA 0.4.0. Discovery pass: `run_discovery.R`. Figures use `nova_theme()/nova_palette()`. Every result shown was selected by effect-size rank from `findings.csv`; none was hand-picked.